



U.S. Department
of Transportation

**Pipeline and Hazardous
Materials Safety
Administration**

April 08, 2025

1200 New Jersey Avenue, SE
Washington, DC 20590

DOT-SP 9370
(THIRTEENTH REVISION)

EXPIRATION DATE: 2029-03-31

(FOR RENEWAL, SEE 49 CFR 107.109)

1. GRANTEE: Norris Cylinder Company
Longview, TX
2. PURPOSE AND LIMITATIONS:
 - a. This special permit authorizes the manufacture, mark, sale, and use of a non-DOT specification cylinder complying in part with the DOT 3T Specification for the transportation in commerce of those hazardous materials authorized in DOT 3T cylinders. This special permit provides no relief from the Hazardous Materials Regulations other than as specifically stated herein. The most recent revision supersedes all previous revisions.
 - b. The safety analyses performed in development of this special permit only considered the hazards and risks associated with transportation in commerce.
 - c. In accordance with 49 CFR 107.107(a), party status may not be granted to a manufacturing special permit. These packagings may be used in accordance with 49 CFR 173.22a.
3. REGULATORY SYSTEM AFFECTED: 49 CFR Parts 106, 107 and 171-180.
4. REGULATIONS FROM WHICH EXEMPTED: 49 CFR §§ 173.301(a)(1) and 173.302(a)(1) in that a non-DOT specification cylinder is not authorized, except as specified herein.
5. BASIS: This special permit is based on Norris Cylinder Company's application dated March 3, 2025, submitted in accordance with § 107.109.

6. HAZARDOUS MATERIALS (49 CFR § 172.101):

Hazardous Materials Description			
Proper Shipping Name	Hazard Class/ Division	Identification Number	Packing Group
Non-liquefied compressed gases, or mixtures of such compressed gases, which are authorized in the Hazardous Materials Regulations to be transported in DOT 3T specification cylinders.	2.1, 2.2, or 2.3, as appropriate	As appropriate	N/A

Note: The following materials are not authorized under this special permit:

- a. Hydrogen, compressed natural gas, hydrogen sulphide, and carbon monoxide;
- b. Any gas mixture containing hydrogen sulfide or other free sulphides, hydrogen, or compressed natural gas;
- c. Any gas mixture containing more than 10% carbon monoxide;
- d. Any gas mixture containing carbon monoxide and having a dew point higher than minus 52 °F at one atmosphere;
- e. Any gas or mixture of gases that does not remain in gaseous state when contained in the cylinder at 70 °F and rated filling pressure; and
- f. Any gas or mixture of gases, the quantity of one or more of which is capable of combining chemically with other gases in such mixture or of combining chemically with the cylinder steel so as to significantly reduce the effectiveness of the cylinder.

7. SAFETY CONTROL MEASURES:

- a. PACKAGING: Packaging prescribed is non-DOT specification steel cylinder manufactured in compliance with DOT 3T (§ 178.45), except as follows:

§ 178.35(c) Duties of the inspector.

[Add]

- (5) Prior to initial production of any design or design change, verify that the design qualification tests prescribed in

this special permit have been performed with acceptable results.

§ 178.35(e) Pressure relief devices and protection for valves and pressure relief devices must comply with § 173.301(f) through (h).

§ 178.45(a) Type, size and service pressure. Each cylinder must be of seamless construction with the bottom convex to pressure. The maximum water capacity is 120 pounds with a minimum service pressure of 1,800 p.s.i.

§ 178.45(b) Material, steel. Only open hearth, basic oxygen, or electric furnace process steel of uniform quality is authorized. The steel analysis must conform to the following:

Analysis Tolerances			
Element (percent)	Ladle analysis	Check analysis	
		Under	Over
Carbon	0.32 to 0.38	0.03	0.04
Manganese	0.60 to 1.05	0.04	0.04
Phosphorus (max)	0.015		0.01
Sulfur (max)	0.010		0.001
Silicon	0.15 to 0.35	0.02	0.03
Chromium	0.80 to 1.15	0.05	0.05
Molybdenum	0.15 to 0.25	0.02	0.02

§ 178.45(c) Manufacture.

[Add]

(6) Cylinder shells must be manufactured by the backward extrusion method.

(7) The thickness of the bottoms of cylinders is, under no condition, to be less than two times the minimum wall thickness of the cylindrical shell; such bottom thickness to be measured within an area bounded by a line

representing the points of contact between the cylinder and floor when the cylinder is in a vertical position.

(8) Each new design and any significant change to any acceptable design must be qualified for production by subjecting at least three prototype samples to pressure cycling tests and burst tests as follows:

(i) Cycle Test. The cycle test must be performed on the completed cylinder after hydrostatic test by subjecting the cylinder to successive hydraulic pressurizations from the lower cyclic pressure to the upper cyclic pressure at a rate not to exceed 10 cycles per minute. Adequate recording instrumentation must be provided if equipment is to be left unattended for any period of time. Lower cyclic pressure must not exceed 10 percent of the upper cyclic pressure. Upper cyclic pressure must be at least equal to the minimum prescribed test pressure.

(ii) Burst Pressure Test. The burst pressure test must be performed on the completed cylinder by hydraulically pressurizing the cylinder to destruction. Rate of pressurization must not exceed 200 psi per second.

(9) In this specification “significant change” means a 10 percent or greater change in cylinder wall thickness, service pressure, or diameter; a 30 percent or greater change in water capacity or base thickness; any change in material; over 100 percent increase in size of openings; or any change in the number of openings.

(10) Ultrasonic Examination: After all shell forming operations and prior to closing in, the cylindrical section of each shell must be examined in accordance with ASTM Standard A-388-80 using the angle beam technique. The equipment used must be calibrated to detect a notch equal to five percent of the design minimum wall thickness. Any discontinuity indication greater than that produced by the five percent notch is cause for rejection of the shell unless the discontinuity is repaired within the requirements of this specification.

§ 178.45(d) Wall thickness. The minimum wall thickness must be such that the wall stress at the minimum specified test pressure does not exceed 67 percent of the minimum tensile strength of the steel as determined by the physical tests required in § 178.45(j) and § 178.45(k). A wall stress of more than 90,500 p.s.i. is not permitted. In no case may wall thickness be less than 0.210 inch.

(1) * * * Except P = minimum test pressure, at least 3/2 service pressure.

(2) Does not apply.

§ 178.45(f) Openings.

(1) Openings are permitted in cylinder head only.

(2) All openings must be threaded. Threads must be in compliance with the following:

(i) Each thread must be clean cut, even, without checks, and to gauge.

(ii) Taper threads, when used, must be in compliance with one of the following:

(A) American Standard Pipe Thread (NPT) type must be in compliance with the requirements of Federal Standard H-28 (1978), Section 7.

(B) National Gas Taper Thread (NGT) type must be in compliance with the requirements of Federal Standard H-28 (1978), Section 7 and 9.

(C) Other taper threads in compliance with other standards may be used provided the length is not less than that specified for NPT threads.

(iii) Straight threads when used must be in compliance with one of the following:

(A) National Gas Straight Thread (NGS) type must be in compliance with the requirements of Federal Standard H-28 (1978), Sections 7 and 9.

(B) Unified Thread (UN) type must be in compliance with the requirements of Federal Standard H-28 (1978), Section 2.

(C) Controlled Radius Root Thread (UNJ) type must be in compliance with the requirements of Federal Standard H-28 (1978) Section 4.

(D) Other straight threads in compliance with other recognized standards may be used provided that the requirements in (2)(iv) below are met.

(iv) All straight threads must have at least 6 engaged threads, a tight fit, and a factor of safety in shear of at least 10 at the test pressure of the cylinder. Shear stress must be calculated by using the appropriate thread shear area per Federal Standard H-28 (1978), Appendix A5, Section 3.

§ 178.45(g) Hydrostatic test. * * *

(6) Each cylinder must be tested to at least 3/2 times its service pressure.

§ 178.45(h) Ultrasonic examination.

[Add]

Wet magnetic particle examination for detecting the presence of quench cracks may be substituted for the ultrasonic examination prescribed in this section. When magnetic particle examination is performed it must be done after the hydrostatic test on the cylindrical section of each cylinder in accordance with ASTM Standard E 709-80. Any cylinder found to have a quenching crack must be rejected and may not be requalified.

§ 178.45(i) Basic requirements for tension and Charpy impact tests.

(1) Two tension specimens and three Charpy impact specimens must be tested from one cylinder taken at random out of each lot of 200 or fewer.

(2) Not applicable.

(3) Each specimen for tension and Charpy impact tests must be taken from the side wall of the cylinder after heat treatment. The axis of the specimens must be parallel to the axis of the cylinder.

(4) The test cylinder need represent only one of the heats in a lot provided the other heats in the lot have been tested and have passed the tests.

(5) Test results must conform to the requirements specified in § 178.45(j) and (k).

(6) When the test results do not conform to the requirements specified, the cylinders represented by the tests may be reheat treated and the tests repeated. 49 CFR 178.45(i)(5) applies to any retesting.

§ 178.45(j) Basic conditions for acceptable physical testing. * * *

(1) * * *

(2) * * *

(3) * * *

(4) Three Charpy impact specimens must be taken from one cylinder in each lot, after heat treatment and tested at -60 °F (-50 °C) or colder. Each impact specimen must be Charpy V-notch type, size 10 mm x 5 mm (1/2 size) prepared and tested in accordance with ASTM Standard E 23-88.

§ 178.45(k) Acceptable results of tests. * * *

(1) * * *

(2) * * *

(3) The Charpy V-notch impact properties for the three 1/2 size (10 mm x 5 mm) impact specimens must not be less than:

Average value for 3 specimens; 25 ft-lb (85 J/cm). Minimum value for any one of the 3 specimens: 21 ft-lb (70 J/cm²).

When the test results do not meet the requirements specified, the lot must be rejected.

(4) * * *

(5) Cylinders subjected to design qualification cycling tests must withstand at least 10,000 cyclic pressurizations without distortion or failure.

(6) Cylinders subjected to design qualification burst tests must withstand a pressure of at least 2.25 times the service pressure without failure. Failure must initiate in the sidewall in a longitudinal direction, and the cylinder must remain in one piece.

§ 178.45(m) Markings. Marking must be in accordance with § 178.35(f) except “DOT-SP 9370” in lieu of “DOT 3T” followed by the service pressure. Rejection elastic expansion (REE) in cubic centimeters(cc) may be marked near the date of test.

b. TESTING: Each cylinder must be requalified for use in accordance with § 180.209 as prescribed for DOT 3T cylinders except that the minimum retest pressure must be 3/2 times the service pressure.

8. SPECIAL PROVISIONS:

a. In accordance with the provisions of Paragraph (b) of § 173.22a, persons may use the packaging authorized by this special permit for the transportation of the hazardous materials specified in paragraph 6, only in conformance with the terms of this special permit.

b. A person who is not a holder of this special permit, but receives a package covered by this special permit, may reoffer it for transportation provided no modifications or changes are made to the package and it is offered for transportation in conformance with this special permit and the HMR.

c. A current copy of this special permit must be maintained at each facility where the package is offered or reoffered for transportation.

d. Each packaging manufactured under the authority of this special permit must be marked with a registration symbol designated by the Office of Hazardous Materials Safety for a specific manufacturing facility.

e. A current copy of this special permit must be maintained at each facility where the package is manufactured under this special permit. It must be made available to a DOT representative upon request.

- f. Reports.
 - (1) Prior to the initial shipment of cylinders made to any specific design, a report of design qualification test results specified in this special permit must be submitted to the Office of Hazardous Materials Safety (OHMS).
 - (2) The manufacturer of the cylinder under this special permit must retain the test reports required by this special permit for 15 years from the original test date on the cylinder.
 - g. These cylinders may not be used for carriage of gases that would cause hydrogen embrittlement of the steel.
 - h. Filling limits specified in § 173.302(c) are not authorized. Under no circumstances are these cylinders to be filled to a pressure exceeding the marked service pressure at 70 °F.
 - i. A copy of the Inspector's report for each of the first three lots produced, must be submitted to the OHMS prior to shipment.
 - j. Cylinders manufactured under this special permit may be transported in an upright position.
 - k. Transportation of Division 2.1 (flammable gases) and Division 2.3 materials (gases, which are poisonous by inhalation) are not authorized aboard cargo vessel or aircraft unless specifically authorized in the Hazardous Materials Table (§ 172.101).
 - l. Transportation of oxygen by aircraft is only authorized when in accordance with § 175.501.
- 9. MODES OF TRANSPORTATION AUTHORIZED: Motor vehicle, rail freight, cargo vessel, and cargo-only aircraft (see restrictions in paragraphs 8.k. and 8.l.).
 - 10. MODAL REQUIREMENTS: A current copy of this special permit must be carried aboard each cargo vessel or aircraft used to transport packages covered by this special permit. The shipper must furnish a current copy of this special permit to the air carrier before or at the time the shipment is tendered.
 - 11. COMPLIANCE: Failure by a person to comply with any of the following may result in suspension or revocation of this special permit and penalties prescribed by the Federal hazardous materials transportation law, 49 U.S.C. 5101 et seq:

- o All terms and conditions prescribed in this special permit and the Hazardous Materials Regulations, 49 CFR Parts 171-180.
- o Persons operating under the terms of this special permit must comply with the security plan requirement in Subpart I of Part 172 of the HMR, when applicable.
- o Registration required by § 107.601 et seq., when applicable.

Each “Hazmat employee”, as defined in § 171.8, who performs a function subject to this special permit must receive training on the requirements and conditions of this special permit in addition to the training required by §§ 172.700 through 172.704.

No person may use or apply this special permit, including display of its number, when this special permit has expired or is otherwise no longer in effect.

Under Title VII of the Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (SAFETEA-LU) — “The Hazardous Materials Safety and Security Reauthorization Act of 2005” (Pub. L. 109-59), 119 Stat. 1144 (August 10, 2005), amended the Federal hazardous materials transportation law by changing the term “exemption” to “special permit” and authorizes a special permit to be granted up to two years for new special permits and up to four years for renewals.

12. REPORTING REQUIREMENTS: Shipments or operations conducted under this special permit are subject to the Hazardous Materials Incident Reporting requirements specified in 49 CFR §§ 171.15 - Immediate notices of certain hazardous materials incidents, and 171.16 - Detailed hazardous materials incident reports. In addition, the grantee(s) of this special permit must notify the Associate Administrator for Hazardous Materials Safety, in writing, of any incident involving a package, shipment or operation conducted under terms of this special permit.

Issued in Washington, D.C.:

A handwritten signature in blue ink, appearing to read 'William Schoonover', is written over a light blue horizontal line.

for William Schoonover
Associate Administrator for Hazardous Materials Safety

Address all inquiries to: Associate Administrator for Hazardous Materials Safety, Pipeline and Hazardous Materials Safety Administration, U.S. Department of Transportation, East Building PHH-13, 1200 New Jersey Avenue, Southeast, Washington, D.C. 20590.

Copies of this special permit may be obtained by accessing the Hazardous Materials Safety Homepage at <https://www.phmsa.dot.gov/approvals-and-permits/hazmat/special-permits-search>. Photo reproductions and legible reductions of this special permit are permitted. Any alteration of this special permit is prohibited.

PO: BB